

Zhijing Wu

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EDUCATION

Sichuan University

B.Eng. in Computer Science

Chengdu, China

Sep. 2025 – Expected Jun. 2030

– Required-course GPA: **4.0/4.0**; Average: **94.7/100**; Rank: **2/288**.

RESEARCH INTERESTS

LLM Agents; Multi-Agent Systems; Agent Communication and Evaluation; Graph-Structured Reasoning; ML Systems for LLMs.

PUBLICATIONS

Jinghong Tang, **Zhijing Wu**, Yikun Liu, Wei Ju, and Jiyao Wang. “Hyperbolic Self-Supervised Pretraining for Hierarchical EEG Representation Learning in Generalizable Parkinson’s Disease Detection.” *2026 IEEE International Conference on Bioinformatics and Biomedicine (IEEE BIBM 2026)*. **Regular Paper, Accepted.**

SELECTED RESEARCH PROJECTS

Communication-Cost Sensitivity in LLM Multi-Agent Protocol Assignment | *Independent Study on ProtocolBench 2026*

- Audited the released ProtocolBench/AgentProtocols GAIA routing path and studied whether workflow communication demand and measured runtime costs can change a fixed local protocol assignment.
- Built an isolated experimental extension without modifying the original benchmark implementation; profiled A2A, ACP, ANP, and Agora over real local request paths across 1–100 KB payloads, with **480/480 measured requests completed successfully**.
- Using measured request costs and official workflow configuration templates, identified an **A2A → ANP** assignment reversal under controlled communication demand; the eligible reversal persisted after globally excluding Agora and separating forced changes.

Execution-Semantics Audit of Guided Topology Diffusion | *Multi-Agent Systems Reproduction Study* 2026

- Reproduced the released GTD pipeline on GSM8K and traced generated communication graphs through diffusion supervision, runtime graph construction, execution, and final aggregation.
- Identified a coordinate mismatch between generated adjacency matrices and runtime edge construction; across **603 records**, raw-to-executed edge Jaccard was **0.152**, while a controlled canonical-coordinate adapter increased it to **0.863**.

Within-Step Corruption Timing for Computer-Use Agents | *Independent Extension of AgentHijack* 2026

- Designed a controlled study of three within-step corruption timings: pre-observation, post-observation/pre-execution, and post-execution, with deterministic perturbation, trajectory logging, and clean-task controls.
- Across **3 Chrome tasks and 36 primary runs**, post-observation/pre-execution corruption increased mean episode length on **all 3/3 tasks (+4.00, +1.75, +1.00 steps)**; one task also showed lower success under stale timing.

Language Modeling Systems from Scratch | *Stanford CS336, Independent Study* 2026

- Implemented core language-model components and training utilities in PyTorch, including BPE tokenization, Transformer modules, optimization, and end-to-end training.
- Studied efficient attention, scaling, and distributed-training systems, including FSDP concepts, together with SFT and RL-based post-training (GRPO/GSPO) for tool-using language models.

AWARDS & HONORS

Meritorious Winner (M Award), Mathematical Contest in Modeling (MCM), 2026.

TECHNICAL SKILLS

Programming: Python, C/C++

Machine Learning: PyTorch, PyTorch Geometric, Transformers, Graph Neural Networks

LLM / Agents: SFT, GRPO/GSPO, Tool-Using Agents, Multi-Agent Systems

ML Systems / Tools: Distributed-training concepts (including FSDP), vLLM, Linux, Git, Docker, Conda